



Right-Triangle Route Planner Teaching Bundle

Grade 8 • North Carolina • Mathematics

NC.8.G.7 • Use the Pythagorean theorem and its converse.

A focused classroom collection: task cards, five warm-ups, classwork, an exit ticket, four seasonal practice sets, substitute lessons, and planning pages. Each section has worked answers.

PRINT • TEACH • REVISIT



Your teaching route

Focus

Use the Pythagorean theorem and its converse. This bundle practices a focused part of NC.8.G.7; it is not a complete grade curriculum or a simulated EOG.

Before teaching

Learners should explain a simpler example using a model. Read the two worked examples, then use the first task card to check readiness.

Suggested sequence

Day 1: model for 10 minutes, then guided task cards for 15 minutes. Day 2: a 5-minute warm-up and 20 minutes of practice. Use the exit ticket to choose a small-group reteaching focus. Seasonal sets revisit the same skill using different quantities.

Materials

Pencils, blank paper, and graph paper or counters as useful. No paid app or internet is needed for student work. A calculator may check finished calculations; these are instructional choices, not official testing accommodation rules.



Make the reasoning visible

Identify the hypotenuse opposite the right angle. Square the legs, add, and take the square root to find the hypotenuse.

Example 1: A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.

Example 2: A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

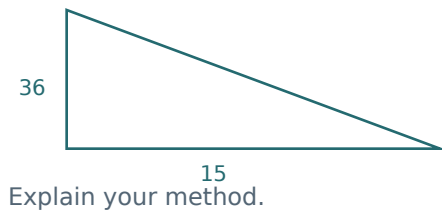
Ask: What does each number represent? How could a model or inverse operation check the result?

Task cards 1-4

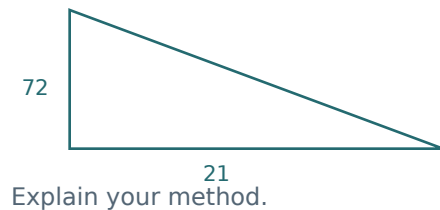
Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 01

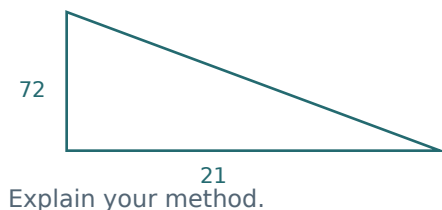
A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 02**

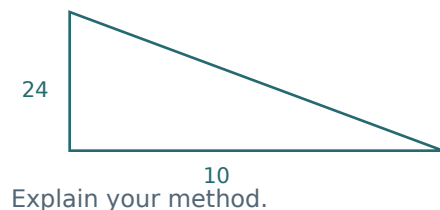
A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 03**

A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 04**

A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

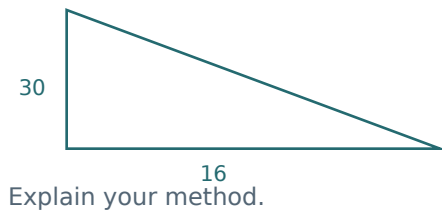


Task cards 5–8

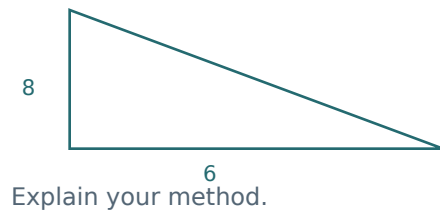
Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 05

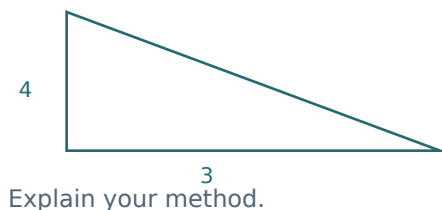
A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 06**

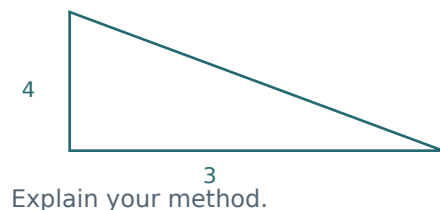
A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 07**

A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 08**

A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

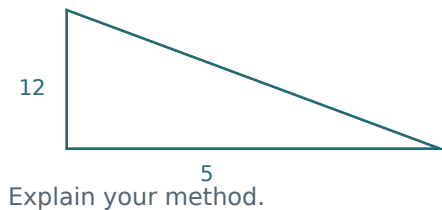


Task cards 9–12

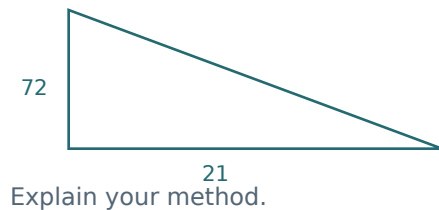
Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 09

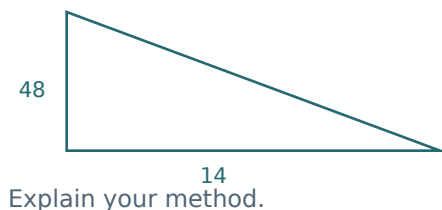
A route goes 5 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 10**

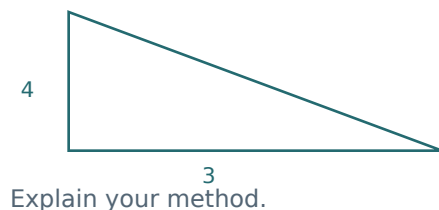
A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 11**

A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 12**

A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

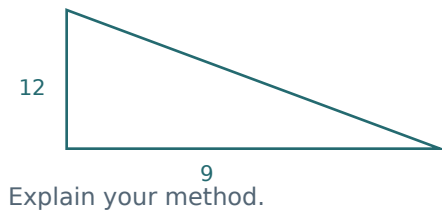


Task cards 13–16

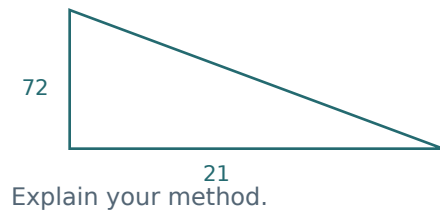
Read, represent, solve, and explain. Record full reasoning on the response page or separate paper.

CARD 13

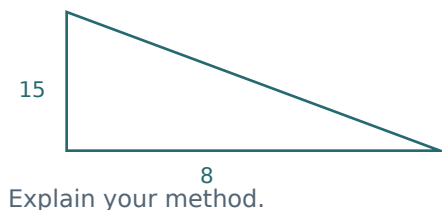
A route goes 9 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 14**

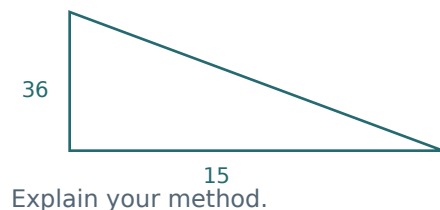
A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 15**

A route goes 8 m east, then 15 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

**CARD 16**

A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Task-card response record

Use this page for final answers; attach drawings or full worked solutions on separate paper.

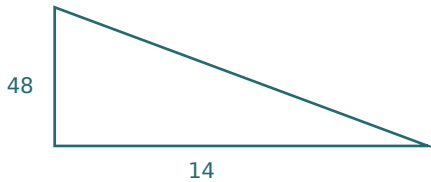
1. _____
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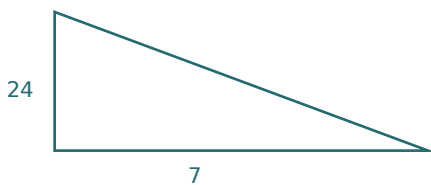
Warm-up • Day 1

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

1. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



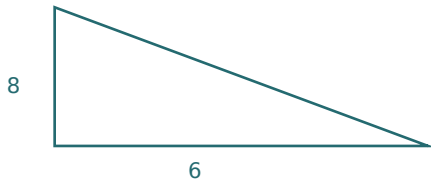
2. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



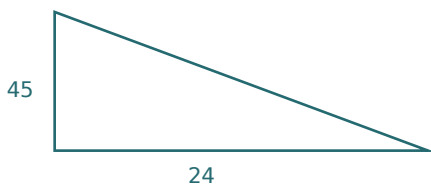
Warm-up • Day 2

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

3. A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



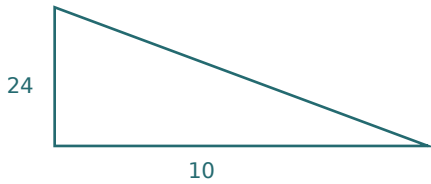
4. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



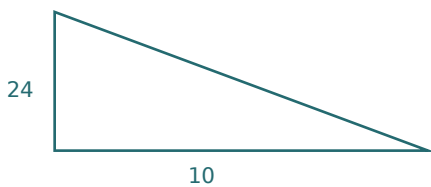
Warm-up • Day 3

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

5. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



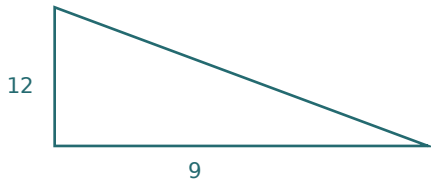
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6. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



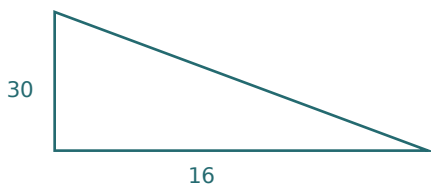
Warm-up • Day 4

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

7. A route goes 9 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



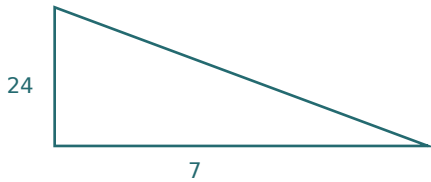
8. A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



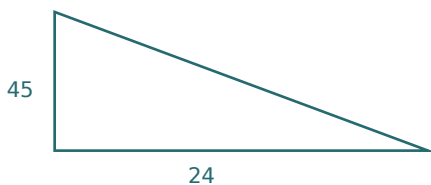
Warm-up • Day 5

Work for about five minutes. Compare one method with a partner. Complete unfinished reasoning during review.

9. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



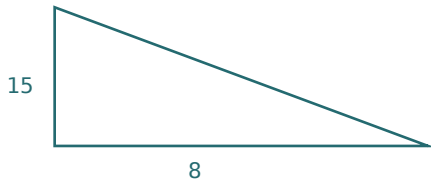
10. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



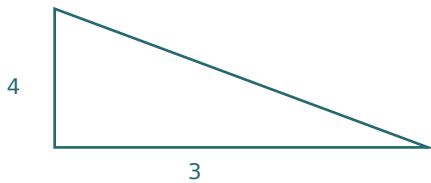
Classwork • Represent and explain

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

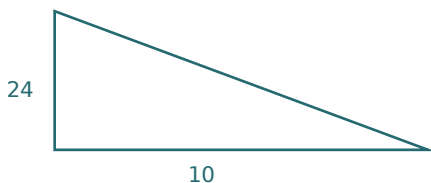
1. A route goes 8 m east, then 15 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



2. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



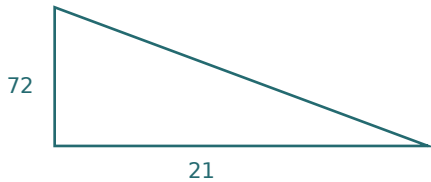
3. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



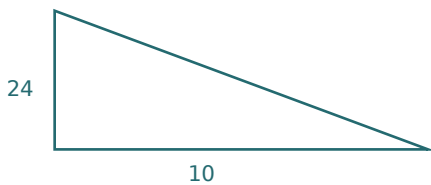
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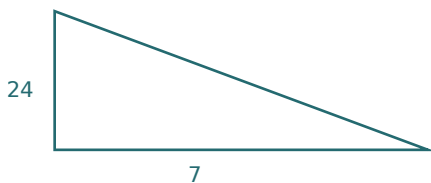
4. A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



5. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



6. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Exit ticket • Try independently

Complete the four questions independently. Show enough reasoning for your teacher to see your method. This is a classroom check, not a standardized score.

1. A route goes 9 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Task-card worked solutions

1. 39 m

The directions form a right angle. $d^2 = 15^2 + 36^2 = 225 + 1296 = 1521$; $d = 39$ m.

2. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441 + 5184 = 5625$; $d = 75$ m.

3. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441 + 5184 = 5625$; $d = 75$ m.

4. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100 + 576 = 676$; $d = 26$ m.

5. 34 m

The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.

6. 10 m

The directions form a right angle. $d^2 = 6^2 + 8^2 = 36 + 64 = 100$; $d = 10$ m.

Task-card worked solutions

7. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

8. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

9. 13 m

The directions form a right angle. $d^2 = 5^2 + 12^2 = 25+144 = 169$; $d = 13$ m.

10. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441+5184 = 5625$; $d = 75$ m.

11. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196+2304 = 2500$; $d = 50$ m.

12. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

Task-card worked solutions

13. 15 m

The directions form a right angle. $d^2 = 9^2 + 12^2 = 81 + 144 = 225$; $d = 15$ m.

14. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441 + 5184 = 5625$; $d = 75$ m.

15. 17 m

The directions form a right angle. $d^2 = 8^2 + 15^2 = 64 + 225 = 289$; $d = 17$ m.

16. 39 m

The directions form a right angle. $d^2 = 15^2 + 36^2 = 225 + 1296 = 1521$; $d = 39$ m.

Warm-up worked solutions

1. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

2. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

3. 10 m

The directions form a right angle. $d^2 = 6^2 + 8^2 = 36 + 64 = 100$; $d = 10$ m.

4. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576 + 2025 = 2601$; $d = 51$ m.

5. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100 + 576 = 676$; $d = 26$ m.

6. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100 + 576 = 676$; $d = 26$ m.

Warm-up worked solutions

7. 15 m

The directions form a right angle. $d^2 = 9^2 + 12^2 = 81 + 144 = 225$; $d = 15$ m.

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The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.

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The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

10. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576 + 2025 = 2601$; $d = 51$ m.

Classwork worked solutions

1. 17 m

The directions form a right angle. $d^2 = 8^2 + 15^2 = 64 + 225 = 289$; $d = 17$ m.

2. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9 + 16 = 25$; $d = 5$ m.

3. 26 m

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6. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

Exit-ticket worked solutions

1. 15 m

The directions form a right angle. $d^2 = 9^2 + 12^2 = 81 + 144 = 225$; $d = 15$ m.

2. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576 + 2025 = 2601$; $d = 51$ m.

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The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.

4. 39 m

The directions form a right angle. $d^2 = 15^2 + 36^2 = 225 + 1296 = 1521$; $d = 39$ m.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Fall • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

6. A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

7. A route goes 5 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

8. A route goes 8 m east, then 15 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Fall • Worked solutions

1. 10 m

The directions form a right angle. $d^2 = 6^2 + 8^2 = 36+64 = 100$; $d = 10$ m.

2. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196+2304 = 2500$; $d = 50$ m.

3. 39 m

The directions form a right angle. $d^2 = 15^2 + 36^2 = 225+1296 = 1521$; $d = 39$ m.

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Fall • Worked solutions

5. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

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The directions form a right angle. $d^2 = 5^2 + 12^2 = 25 + 144 = 169$; $d = 13$ m.

8. 17 m

The directions form a right angle. $d^2 = 8^2 + 15^2 = 64 + 225 = 289$; $d = 17$ m.

Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A route goes 6 m east, then 8 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Winter • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

6. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

7. A route goes 5 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

8. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Winter • Worked solutions

1. 10 m

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2. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441+5184 = 5625$; $d = 75$ m.

3. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576+2025 = 2601$; $d = 51$ m.

4. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100+576 = 676$; $d = 26$ m.

Winter • Worked solutions

5. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

6. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9 + 16 = 25$; $d = 5$ m.

7. 13 m

The directions form a right angle. $d^2 = 5^2 + 12^2 = 25 + 144 = 169$; $d = 13$ m.

8. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9 + 16 = 25$; $d = 5$ m.

Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 16 m east, then 30 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Spring break • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

6. A route goes 15 m east, then 36 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

7. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

8. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Spring break • Worked solutions

1. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576 + 2025 = 2601$; $d = 51$ m.

2. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

3. 34 m

The directions form a right angle. $d^2 = 16^2 + 30^2 = 256 + 900 = 1156$; $d = 34$ m.

4. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.



Spring break • Worked solutions

5. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

6. 39 m

The directions form a right angle. $d^2 = 15^2 + 36^2 = 225 + 1296 = 1521$; $d = 39$ m.

7. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

8. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100 + 576 = 676$; $d = 26$ m.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

1. A route goes 8 m east, then 15 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 9 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 5 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Summer • Skill revisit

Complete four questions per sitting, about 15–20 minutes. Review the worked examples if needed. Explain one answer to a classmate or adult.

5. A route goes 5 m east, then 12 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

6. A route goes 24 m east, then 45 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

7. A route goes 7 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

8. A route goes 21 m east, then 72 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Summer • Worked solutions

1. 17 m

The directions form a right angle. $d^2 = 8^2 + 15^2 = 64 + 225 = 289$; $d = 17$ m.

2. 15 m

The directions form a right angle. $d^2 = 9^2 + 12^2 = 81 + 144 = 225$; $d = 15$ m.

3. 13 m

The directions form a right angle. $d^2 = 5^2 + 12^2 = 25 + 144 = 169$; $d = 13$ m.

4. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9 + 16 = 25$; $d = 5$ m.

Summer • Worked solutions

5. 13 m

The directions form a right angle. $d^2 = 5^2 + 12^2 = 25 + 144 = 169$; $d = 13$ m.

6. 51 m

The directions form a right angle. $d^2 = 24^2 + 45^2 = 576 + 2025 = 2601$; $d = 51$ m.

7. 25 m

The directions form a right angle. $d^2 = 7^2 + 24^2 = 49 + 576 = 625$; $d = 25$ m.

8. 75 m

The directions form a right angle. $d^2 = 21^2 + 72^2 = 441 + 5184 = 5625$; $d = 75$ m.

Substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use the Pythagorean theorem and its converse. Let students explain a familiar strategy to a partner.

5-15 minutes: model

Use the worked-example page. Identify the hypotenuse opposite the right angle. Square the legs, add, and take the square root to find the hypotenuse.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A route goes 10 m east, then 24 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 8 m east, then 15 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 14 m east, then 48 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Substitute lesson • Worked solutions

1. 26 m

The directions form a right angle. $d^2 = 10^2 + 24^2 = 100 + 576 = 676$; $d = 26$ m.

2. 17 m

The directions form a right angle. $d^2 = 8^2 + 15^2 = 64 + 225 = 289$; $d = 17$ m.

3. 50 m

The directions form a right angle. $d^2 = 14^2 + 48^2 = 196 + 2304 = 2500$; $d = 50$ m.

4. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9 + 16 = 25$; $d = 5$ m.

Emergency substitute lesson

Before students arrive

Locate the class schedule, seating plan, emergency procedures, and approved learner supports from the school. This academic packet does not replace those school-specific instructions.

0-5 minutes: settle

Read the goal aloud: Use the Pythagorean theorem and its converse. Let students explain a familiar strategy to a partner.

5-15 minutes: model

No copies or devices? Write the first worked example on the board; students use blank paper. Identify the hypotenuse opposite the right angle. Square the legs, add, and take the square root to find the hypotenuse.

15-30 minutes: practice

Distribute the next practice page. Students solve independently first, then compare one explanation in pairs. If copying is unavailable, read one question at a time and allow drawings or oral responses.

30-40 minutes: discuss

Use the following key. Ask students to locate the first step where methods diverged; invite a correction supported by a model.

40-45 minutes: handover

Collect one explanation from each student. On the planning page, record which step needs reteaching, who requested help, and what was completed. Use neutral observations, not diagnostic labels.

Emergency substitute lesson • Practice

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

2. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

3. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.

4. A route goes 3 m east, then 4 m north. Find the direct straight-line distance from start to finish. Explain why the theorem applies.



Emergency substitute lesson • Worked solutions

1. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

2. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

3. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

4. 5 m

The directions form a right angle. $d^2 = 3^2 + 4^2 = 9+16 = 25$; $d = 5$ m.

A completed lesson-plan example

Level and focus

Grade 8; NC.8.G.7; Use the Pythagorean theorem and its converse.

Evidence of learning

Student represents the quantities, selects a valid method, calculates accurately, and explains the result with appropriate units.

Launch / model

Use worked example 1. Ask students to annotate the role of every quantity. Demonstrate the model before moving to symbols.

Guided / independent practice

Use cards 1-4 in pairs; assign classwork independently. Change grouping according to the explanation you hear rather than speed alone.

Formative check

Use the four-question exit ticket. Score each item 0-2: 2 for an accurate result with coherent reasoning, 1 for a sound method with a minor error or an unexplained correct result, 0 for missing or unsupported reasoning.

Next teaching decision

0-3 points: reteach with a concrete model. 4-6: correct the specific misconception with a short example. 7-8: invite a second representation or a student-created problem. These are local instructional decisions, not mastery cutoffs validated by a test publisher.

Plan your next lesson

Date, class, and focus / exact local standard:

What will students do to show understanding?

Launch and worked example:

Guided practice and independent task:

Access supports and extension:

Exit evidence and next lesson decision:



Access, sources, and classroom license

AIG: Invite two methods, a counterexample, or a new task with a specified solution. Ask students to explain limits of a shortcut.

504: Apply the student's plan: breaks, extra time, enlarged print, reduced copying, or alternate response methods as documented.

EC: Use the IEP to select supports. Model one step, offer counters or a diagram, reduce the visible item set, and accept an oral explanation when appropriate.

ESL: Preview quantity, difference, factor, or other relevant terms. Let students rehearse in a home language, then use "I chose ___ because ___." Maintain the mathematical goal.

Official alignment sources • accessed September 15, 2026

NC.8.G.7 is verified in the NCDPI mathematics progression. The activity objective is original wording; it describes the skill focus, not the full standard text.

NCDPI mathematics standards:

<https://www.dpi.nc.gov/documents/publications/catalog/ma196-vertical-progression-math/open>

NCDPI EOG mathematics specifications:

<https://www.dpi.nc.gov/documents/accountability/testing/eog/eog-mathematics-grades-3-8-test-specifications>

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